Applied and Computational Mathematics
2025; 9 de March: 30 July
https://www.sciencepg.com/article/10.11648/j.acm.20200903.15
DOI: https://doi.org/10.5281/zenodo.16421895
ISSN: xxxx-xxxx (Print); ISSN: xxxx-xxxx (Online)

Heuristic Algorithms for Generalized Vehicle Routing Problem (H-GVRP)

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To cite this article:

Hernández Gómez, F. J. & Palacios López W. J. (2025). Heuristic Algorithms for Generalized Vehicle Routing Problem (H-GVRP). *Applied and Computational Mathematics, Volume*(4), Page Range: 1- 11. DOI: https://doi.org/10.5281/zenodo.16421895

Received: 9 March 2025; **Accepted:** 29 April 2025; **Published:** 30 July 2025

Abstract:

The Generalized Vehicle Routing Problem (GVRP) represents a significant advancement in Operations Research, emerging as a foundational element in the optimization of logistics and transportation systems. As an extension of the classic Vehicle Routing Problem (VRP), the GVRP is uniquely characterized by its ability to cluster customers, enabling a single vehicle to efficiently serve multiple clients within designated groups. This clustering capability substantially enhances route efficiency and reduces operational costs. To tackle the complexities of the GVRP, a variety of heuristic algorithms have been developed, integrating Metaheuristic approaches such as ant colony optimization and genetic algorithms. These methodologies are particularly effective in managing heterogeneous fleets with diverse capacities and varied delivery requirements, thus ensuring flexibility and adaptability in route planning. The integration of the Internet of Things further enhances this process by facilitating seamless communication between vehicles and management systems, leading to improved operational efficiency. In light of increasing demands for sustainability, the GVRP has evolved towards more innovative approaches, solidifying its importance in contemporary logistics research. This evolution not only responds to the need for effective and efficient routing solutions but also emphasizes the necessity of incorporating sustainable practices in route planning and execution. Consequently, the GVRP continues to be a dynamic area of research, driving both optimization advancements and sustainable transportation strategies.

Keywords: Algorithms, Heuristics, Optimization, Vehicle Routing, GVRP

1. Introduction

This paper explores recent advancements concerning the Generalized Vehicle Routing Problem (GVRP), underscoring its critical relevance in the logistics and transportation sector. Since the inception of the Vehicle Routing Problem (VRP) in 1959, extensive research has been dedicated to its exploration, leading to significant evolutionary developments in the field [6, 7, 3, 4]. The ongoing collaboration between academic researchers and industry practitioners is vital for maintaining the VRP's relevance in optimizing resources and improving logistics operations. Such cooperative efforts foster the development of more precise optimization models and enable the validation of solutions in practical settings.

This collaborative approach not only enhances the rigor and applicability of research findings but also ensures that the VRP continues to evolve in response to real-world challenges, driving advancements that align with the industry's growing demand for efficiency and sustainability.

1.1. Problem Statement

The Generalized Vehicle Routing Problem (GVRP) presents a substantial challenge within Operations Research and Optimization field Combinatorial, reflecting its significant relevance across a multitude of practical applications. At its core, the GVRP aims to optimize vehicle routes to minimize costs while serving a set of geographically dispersed customers. To achieve this goal, a fleet of vehicles must depart from and return to designated depots, ensuring that all assigned customers are visited within specified operational constraints.

The GVRP is distinguished by its numerous adaptations tailored to the unique operational characteristics of different companies. These adaptations may encompass various factors, including vehicle capacity, frequency of visits, types of deliveries, and specific delivery schedules and conditions. Such diverse requirements highlight the necessity for customized routing solutions that effectively address the distinct needs of various logistical environments.

To tackle the complexities of the GVRP, a wide range of techniques and methodologies have been proposed. While some approaches employ exact optimization methods—such as integer programming and constraint programming—these often face challenges related to computational efficiency, particularly under realistic conditions. Conversely, heuristic and meta-heuristic strategies have proven effective in generating high-quality solutions within reasonable computational times, albeit without formal guarantees of optimality.

In recent years, innovative methods have emerged that integrate combinatorial optimization techniques with heuristic algorithms, such as genetic algorithms, ant colony optimization, tabu search, and simulated annealing [9, 19, 15, 5]. These hybrid approaches have shown promising results, significantly improving both solution quality and computational efficiency. The exploration of such integrated methodologies not only addresses the multifaceted challenges associated with the Generalized Vehicle Routing Problem (GVRP) but also paves the way for more efficient and sustainable logistics solutions.

1.2. Research Hypothesis

The primary hypothesis of this research focuses on the computational complexity associated with the Generalized Vehicle Routing Problem (GVRP) when utilizing heterogeneous fleets. This complexity presents significant challenges that hinder the effective application of exact methods within a reasonable time frame. Consequently, this study advocates for the use of heuristic and meta-heuristic algorithms based on iterative approaches to efficiently identify solutions that minimize system costs in a practical manner.

To validate this hypothesis, the research will investigate innovative strategies, including the integration of meta-heuristic techniques, real-time adaptive strategies, and multi-objective optimization. Furthermore, the study will explore the potential application of machine learning algorithms to model customer demand patterns, with the aim of improving route planning and vehicle allocation processes.

The overarching objective of this research is to enhance the efficiency of vehicle routing operations, reduce associated costs, and mitigate environmental impacts. It is anticipated that the implementation of these methodologies will yield high-quality solutions for the GVRP within a reasonable computational time frame, thereby overcoming the limitations of traditional exact methods and achieving promising results in both optimization and sustainability.

1.3. Justification of the Problem

The Vehicle Routing Problem (VRP), including its variants such as the Generalized Vehicle Routing Problem (GVRP), is fundamental in combinatorial optimization and has significant applications across various industries, including product distribution and passenger transportation. These problems offer multiple approaches to optimize transport logistics, effectively reducing costs and enhancing operational efficiency. Efficient route planning and proper vehicle allocation can lead to substantial savings in transportation costs, which can account for up to 20% of the final product costs. Research has demonstrated that effective routing strategies can reduce these costs by 5% to 20% by improving vehicle utilization and minimizing total distances traveled.

Various strategies exist within the Vehicle Routing Problem (VRP) framework, each tailored to address specific logistical challenges. These strategies have been extensively explored in recent literature [19, 3, 12, 16]. Recent studies indicate that hybrid approaches and variable neighborhood search methodologies can outperform traditional methods regarding both solution quality and execution time, particularly in vehicle routing problems involving heterogeneous fleets [13, 18, 20]. Additionally, strategies that incorporate real-time data, resource sharing, and environmental considerations are essential for developing more effective and sustainable distribution networks. These advancements play a pivotal role in improving logistical efficiency and reducing transportation costs.

Heterogeneity in vehicle fleets is a common characteristic among companies, manifesting as variations in capacities such as load, volume, and refrigeration needs. By acknowledging this diversity, the proposed heuristic algorithms can be better tailored to meet operational requirements, yielding optimal and practical solutions. The significance of addressing the GVRP lies in the economic and environmental benefits that can be achieved through optimizing route planning and vehicle allocation, thereby reducing costs and minimizing carbon emissions. Consequently, the development of efficient and effective algorithms for this problem is critical for achieving positive outcomes in logistics management.

Furthermore, several key points underscore the importance of studying such problems:

1.3.1. Economic and Environmental Importance

The Generalized Vehicle Routing Problem (GVRP) and its variants are essential in various industries, including product distribution, waste collection, and transportation for individuals with disabilities. Optimizing route planning and vehicle allocation not only results in significant savings in transportation costs, which account for a substantial portion of the final product cost, but also positively influences carbon emission reduction and environmental preservation. This is achieved by decreasing the total distance traveled and enhancing the efficiency of vehicle utilization [20].

1.3.2. Limitations of Existing Approaches

Despite the efforts in research and proposals to address the GVRP, it is essential to acknowledge the limitations and challenges that persist. Some current approaches may present prolonged execution times or fail to achieve the desired efficiency in solution quality. Consequently, there is a justified need to explore new approaches or models, such as the heuristic and meta-heuristic algorithms proposed in this research project, which have the potential to provide higher quality solutions in a reasonable time frame.

1.3.3. Heterogeneity of the Vehicle Fleet

In today's business environment, it is common to encounter vehicle fleets that exhibit a variety of characteristics, such as load capacity, volume, and refrigeration transport needs, among others. Considering this diversity in route planning and vehicle allocation allows for precise adaptation to the operational and logistical demands of companies, resulting in optimal and realistic solutions. This diversity adds complexity to the Generalized Vehicle Routing Problem (GVRP), justifying the need to develop heuristic algorithms tailored to this particularity in order to effectively address the needs and constraints arising from fleet heterogeneity [9, 18, 5].

1.3.4. Incorporation of Novel and Creative Approaches

The integration of meta-heuristic techniques with approaches derived from artificial intelligence, multi-objective optimization, game theory, and machine learning significantly expands the possibilities for innovatively addressing the Generalized Vehicle Routing Problem (GVRP). These strategies allow for improved solution quality by considering key aspects such as vehicle efficiency, carbon emission reduction, and customer satisfaction. The implementation of dynamic and adaptive approaches enables real-time adjustments in route planning and vehicle allocation, allowing for effective responses to changes in demand or unforeseen events [2].

1.3.5. Novel and Creative Contributions

The proposal to utilize heuristic algorithms based on a method of successive approximations to solve the GVRP with a heterogeneous fleet constitutes a novel and creative contribution. These algorithms have the potential to provide high-quality solutions in a reasonable time frame, thereby allowing companies to optimize their routing operations and improve logistical efficiency. The consideration of fleet heterogeneity adds complexity to the problem, necessitating specialized approaches to accurately model vehicle constraints and capabilities, thereby generating realistic and feasible solutions.

Furthermore, the combination of meta-heuristic techniques and multi-objective optimization broadens the avenues for tackling the GVRP [16], exploring new ways to enhance solution quality by considering factors such as vehicle efficiency, carbon emissions, and customer satisfaction. The application of approaches based on game theory and machine learning facilitates the incorporation of dynamic and adaptive strategies, intelligently and efficiently adjusting route planning and vehicle allocation in response to real-time changes.

1.4. Research Objectives

In the framework of the Heuristic Algorithms for Generalized Vehicle Routing Problem (AH-GVRP), a overarching objective, along with several specific aims, has been formulated to steer this research. The primary objective encompasses the development of efficient algorithms designed to optimize the GVRP. To accomplish this overarching aim, a series of specific objectives have been delineated, which are elaborated upon below.

Firstly, a meticulous examination of the current state of research concerning vehicle routing problems will be undertaken. This includes a detailed exploration of foundational variants such as the Traveling Salesman Problem (TSP), the multi-Transportation Problem (m-TSP), the Vehicle Routing Problem (VRP), and the GVRP itself. This analytical process aims to identify and synthesize innovative methodologies and techniques previously utilized to address these challenges, thereby enhancing the optimization of route planning and vehicle assignment in practical applications.

Secondly, the research aims to design effective algorithms for the AH-GVRP, with particular attention to the heterogeneity present in vehicle fleets. These algorithms will employ successive approximation strategies, global search methodologies, and comprehensive solution space exploration techniques to significantly improve the efficiency of goods distribution while concurrently minimizing transportation costs.

In addition, the effectiveness of the proposed AH-GVRP algorithms will be rigorously evaluated in comparison to other existing solutions. This evaluation will utilize extensive computational experiments along with relevant performance metrics, facilitating a robust determination of the algorithms' ability to produce high quality solutions and their practical feasibility in contemporary applications.

Finally, the effectiveness of the AH-GVRP algorithms will be validated through their application in practical cases. The subsequent evaluation will analyze their impact on reducing transportation costs, optimizing resource utilization and improving service quality.

2. Theoretical Framework

2.1. State of Art

The earliest studies on the Vehicle Routing Problem (VRP) trace back to 1959, when Dantzig and Ramser developed the first mathematical model aimed at optimizing the distribution of gasoline across various service stations. This model effectively illustrates the challenges associated with distribution logistics. A significant contribution to the field was made by Golden et al. [8], who proposed a generalization that integrates the VRP with the Traveling Salesman Problem (TSP). This integration facilitates the solution of vehicle allocation and the determination of customer visit sequences. Over the years, more recent developments have shifted focus toward more flexible and adaptive approaches, incorporating environmental and sustainability considerations into route planning.

In addition, Kara and Bektas [11] introduced the concept of the Generalized Vehicle Routing Problem (GVRP), which includes multiple depots and heterogeneous demands. This generalization has allowed for the analysis of more complex and realistic logistics scenarios. The integration of emerging technologies has enabled more efficient collaboration between vehicles and facilitated intelligent decision-making through real-time communication. As a result, route design has been optimized, and unnecessary congestion has been alleviated, thereby enhancing the efficiency of goods delivery and reducing operating costs.

Traditionally, the resolution of the Vehicle Routing Problem (VRP) has relied on heuristic and meta-heuristic algorithms, such as Tabu search and genetic algorithms. However, recent advancements have unlocked the potential of advanced optimization techniques based on artificial intelligence, including evolutionary algorithms, neural networks, and reinforcement learning. These innovations are capable of addressing more complex challenges and producing high-quality solutions in significantly reduced time frames.

The proposed generalizations, including the Generalized Vehicle Routing Problem (GVRP), coupled with the integration of autonomous vehicles and advanced optimization techniques, have facilitated the development of more efficient and sustainable solutions. This advancement is supported by the work of Golden and Assad [8] and Kara and Bektas [11]. These innovative methodologies significantly impact logistics and transportation by reducing delivery times, optimizing resource utilization, and contributing to lower emissions and energy consumption, as discussed by Penna et al. [18].

Moreover, the combination of machine learning techniques with data-driven optimization methods promises to elevate the efficiency and quality of solutions to unprecedented levels, according to Liu et al. [14]. The application of real-time data, the incorporation of emerging technologies, and optimization strategies that integrate sustainability criteria further lay the groundwork for intelligent decision-making and continuous improvement in vehicle routing.

In the Vehicle Routing Capacity Problem (CVRP), the main objective is to determine the optimal number of routes that minimize the total cost. This cost is defined as the sum of the costs of the arcs that make up the routes. It is necessary for each route to include a visit to the distribution center, and each consumption center must be served by a single vehicle. In addition, the sum of the demands of the consumption centers visited on each route cannot exceed the capacity of the vehicle used. It is important to note that there are two variants of CVRP depending on the symmetry of transportation costs. In the symmetric CVRP, the cost of moving from one consumption center i to another consumption center j is equal to the cost of doing so in the opposite direction, i.e., from j to i. In contrast, in asymmetric CVRP, transportation costs may differ depending on the direction of travel. This distinction is relevant, since it has significant implications in the formulation and resolution of the problem, allowing the model to be adapted to specific situations depending on the nature of the transportation costs between consumption centers.

2.2. Modelo Matemático para el (GVRP)

In the subsequent section, we derive the mathematical model for the Generalized Vehicle Routing Problem (GVRP), with the primary objective of minimizing the total cost associated with the routes taken by the vehicles. The model considers a scenario with a predetermined maximum number of vehicles, each having a fixed homogeneous capacity. These vehicles commence their journeys from a designated point known as the depot, which serves as the origin for all routes aimed at servicing a set of geographically dispersed customers. Each customer location has a specific and known demand, which must be fulfilled by the single vehicle assigned to visit that location.

To effectively represent this problem, we introduce indices and variables that will allow us to formulate the mathematical framework necessary for analyzing and optimizing the vehicle routes. The following notation will be utilized:

- i: index representing the starting node of a vehicle.
- j: index representing the destination node of a vehicle.
- k: vehicle to use $(1, 2, 3, \ldots, k)$.
- d_i: demand of the customer located at node i.
- x_{ij}^k: equals 1 if vehicle k is assigned to traverse the arc from node i to node j; equals 0 otherwise.
- y_{ij}^k: equals 1 if the traversal from node i to node j occurs; equals 0 otherwise.

The mathematical routing model is based on the foundational contributions of Ahuja, R, and et. all. [1]. In their work Network Flows: Theory, Algorithms, and Applications, these authors present a theoretical and algorithmic framework that thoroughly addresses routing in networks. Furthermore, highlight the significance of implementing efficient algorithms for solving the Generalized Vehicle Routing Problem (GVRP)

[10]. Their research includes the proposal of various heuristics specifically designed to tackle routing challenges in the vehicle transportation sector. These texts are essential references for the development and application of mathematical models across various applications in the field of routing.

$$\text{Minimize } z(x) = \sum_{(i,j)\in A} c_{ij} \cdot y_{ij} \tag{1}$$

In this model, the objective function in (2.2.1) indicates that the goal is to minimize the sum of the costs of traveling from the depot to all nodes, where the cost is primarily based on the distance between those nodes.

Subject to:

$$\sum_{1\le k\le K} x_{ij}^k = y_{ij}, \quad \forall i,j \tag{2}$$

$$\sum_{1\le i\le n} y_{ij} = 1, \quad \forall j \tag{3}$$

$$\sum_{1\le j\le n} y_{ij} = 1, \quad \forall i \tag{4}$$

$$\sum_{1\le j\le n} y_{0j} = k \tag{5}$$

$$\sum_{1\le j\le n} y_{i0} = k \tag{6}$$

$$\sum_{1\le i\le n} \sum_{1\le j\le n} d_i \cdot x_{ij}^k \le u, \quad \forall k \tag{7}$$

$$\sum_{i\in Q} \sum_{j\in Q} y_{ij} \le |Q| - 1, \quad \forall \text{ subset of} \tag{8}$$

$$Q \text{ from } \{1,2,\ldots,n\} \tag{9}$$

$$k \le K \tag{10}$$

$$y_{ij} \in \{0,1\}, \quad \forall (i,j) \in A \tag{11}$$

$$x_{ij}^k \in \{0,1\}, \quad \forall (i,j) \in A, \forall k. \tag{12}$$

Where, the set A is defined as $A = \{(i,j) : y_{ij} = 1\}$. That is, the set of edges of the graph allowing for a single traversal from node i to node j. All routes start at 0 and terminate at 0. Each arc $(i,j) \in A$, with $i \neq j$, in the network is associated with a cost c_{ij}. Each vehicle is characterized by having a load capacity q. Each customer presents a demand d_i, with $i \in C$. Moreover, it is assumed that all data for each of the customers are known non-negative integers. The model is deterministic, and the constraints presented in the model aim to design a set of minimum-cost routes, one for each vehicle, such that:

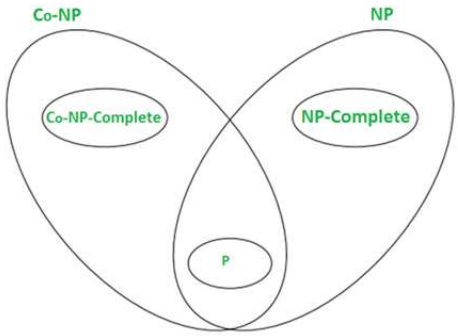
- Each customer is served exactly once.
- Each route starts and ends at the depot.
- The capacity restrictions of the vehicles are respected.

2.3. Computational complexity

The exhaustive method for solving the Generalized Vehicle Routing Problem (GVRP) entails an extremely high computational cost, thereby limiting its applicability in real-time environments and dynamic situations . This challenge has significantly driven the development of heuristic algorithms, which are notably faster and often yield satisfactory quality solutions within reasonable time frames. Such approaches are crucial given the interplay of multiple factors inherent to the GVRP, including the heterogeneity of vehicle fleets and variations in demand [17].

In the realm of computational complexity, decision problems like the Generalized Vehicle Routing Problem (GVRP) are categorized into several classes, including L, NL, P, NP, NP-Complete, NP-Hard, and $Co\text{-}NP$, according to the theoretical framework of Turing machines (TM). The GVRP falls under the NP classification, highlighting that no polynomial-time algorithm is known to solve all instances of this problem efficiently. Figure 1 provides a visual summary of these complexity classes.

Figure 1. *Classifications of the computational complexity of combinatorial optimization problems.*



Note: *This figure showcases the computational complexity classes associated with the Generalized Vehicle Routing Problem (GVRP).*

Fascinating questions arise regarding the relationships among these classes, such as whether

$$P = NP, Co - NP = NP, \text{ and } P = Co - NP \cap NP.$$

Some of these questions remain unresolved, continuing to be subjects of research for computer scientists, theoretical mathematicians, and computational scientists. Resolving such issues could revolutionize the field of combinatorial optimization and yield new strategies for addressing complex problems like GVRP.

Given the complexity of the GVRP, the use of heuristics and meta-heuristics becomes increasingly relevant as they balance optimization with temporal efficiency. Recent studies suggest that algorithms combining global search approaches with adaptive techniques often outperform traditional heuristics, providing higher-quality solutions without compromising computational performance.

Furthermore, the integration of machine learning techniques with optimization algorithms presents a promising avenue for managing GVRP complexity. These innovations not only enhance efficiency in route planning but also facilitate the development of adaptive routing systems responsive to

dynamic environmental conditions, marking a significant advancement in operations research and logistics.

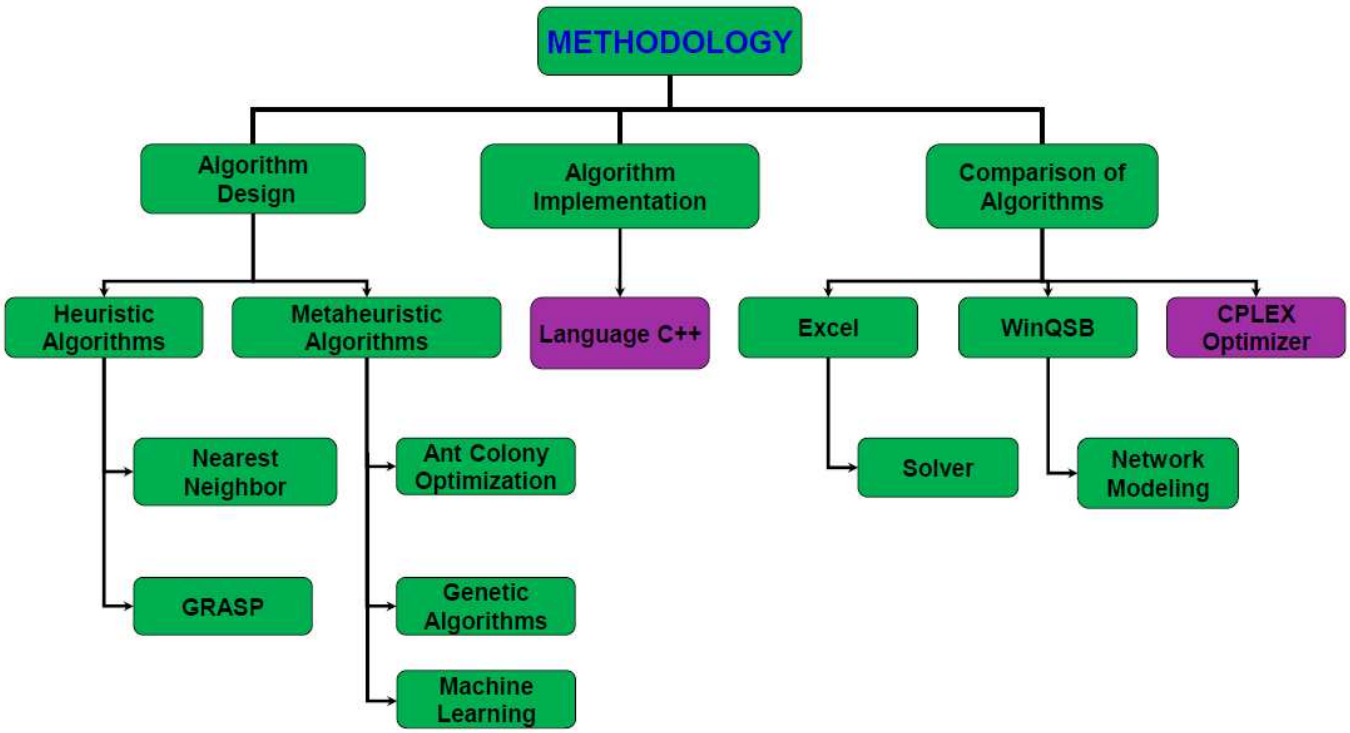
3. Applied Methodology

The proposed study is an applied research endeavor focused on developing and enhancing algorithms. Applied research aims to create practical and effective solutions for real-world challenges. This study specifically targets the design and development of efficient algorithms, including both heuristics and meta-heuristics, that are well-regarded in the scientific community for addressing the Generalized Vehicle Routing Problem in real-world contexts. Furthermore, the heuristic algorithms developed for the GVRP offer efficient and practical solutions, successfully balancing computational costs with solution quality. These methodologies are particularly essential for real-time applications, as they provide the necessary flexibility and adaptability to accommodate various constraints and dynamic conditions inherent to the GVRP.

The research approach is both quantitative and qualitative, involving the collection of numerical and qualitative data, along with statistical analyses and case studies to assess the effectiveness of the proposed algorithms. We investigated variations in parameters and implemented advanced operators within the meta-heuristic models, resulting in the development of hybrid approaches. Moreover, a synthesis analysis method was applied to decompose the problem into more manageable sub-problems and to integrate the results from various algorithms, thereby enhancing the overall outcomes.

To effectively demonstrate and validate the algorithmic results, abstraction and concretion techniques were utilized. Notably, the algorithms were implemented in the C++ programming language, and IBM ILOG CPLEX Optimization Studio was employed to optimize performance during computational laboratory experiments, leveraging the strengths of these methodologies. Figure 2 illustrates the methodological framework of the proposed study.

Figure 2. *Framework of the methodology used*



Note: *This figure presents the methodological framework employed in this study, highlighting the computational complexity classes to the Generalized Vehicle Routing Problem (GVRP). Therefore, it serves to illustrate the structured approach taken in developing and analyzing the problem's various aspects.*

Similarly, we can highlight other relevant aspects of the methodological design, which are shown below.

3.1. General Objective

To develop and validate algorithms designed for the generalized vehicle routing problem (GVRP) in real-world applications that facilitate the optimization of route planning and vehicle assignment. This highlights the intention to address the problem from various approaches and to explore optimal solutions.

3.2. Approach

A mixed-method approach is utilized in the research, integrating both quantitative and qualitative data collection

at each stage of the study and evaluation of the proposed algorithms. Numerical data from real cases will be gathered, and statistical analyses will be conducted to assess the performance of the algorithms from a quantitative perspective. Additionally, qualitative information will be collected through case studies and feedback from users and experts in the field, which will enrich the evaluation of the algorithms and provide additional insights into their effectiveness in real-world applications.

3.3. Techniques and Materials

Techniques for designing proposed heuristic and metaheuristic algorithms are employed; other approaches, such as machine learning or multi-objective optimization, will be explored with the intent to leverage the heterogeneity of the vehicle fleet by applying successive approximations, as well as global search techniques and exploration of the solution space. The proposed algorithms will be implemented modularly and efficiently using an appropriate programming language. Furthermore, data visualization techniques will be considered to graphically represent the solutions found by the developed algorithms, facilitating the interpretation of results and their communication more effectively.

3.3.1. Data Collection

Existing sources for compiling GVRP datasets are mentioned. As an additional contribution, the possibility of generating synthetic datasets representing different scenarios and conditions is suggested, which will allow for an expanded diversity of test cases and validate the robustness of the proposed algorithms.

3.3.2. Development of the AH-GVRP Algorithms

Efficient algorithms based on heuristics and metaheuristics for the generalized vehicle routing problem will be designed and implemented. As an additional contribution, consideration could be given to the implementation of parameter optimization techniques to adjust the parameters of the algorithms and enhance their performance across different instances of the problem.

3.3.3. Evaluation of the Proposed Algorithms

Computational experiments will be conducted using real and synthetic datasets to evaluate the effectiveness of the proposed algorithms. The results obtained will be compared with existing approaches using relevant metrics, such as total transportation cost, resource utilization, and service quality. Moreover, the inclusion of additional metrics related to the robustness of the found solutions could be considered, such as sensitivity to changes in data or adaptability to dynamic scenarios.

3.3.4. Results Analysis

The proposed statistical analysis is valid and will allow for the evaluation of the algorithms' effectiveness in terms of efficiency and quality of the solutions found. Additionally, a qualitative analysis of the results is suggested, such as examining specific case studies or identifying emerging patterns in the solutions, which will provide a more comprehensive understanding of the obtained results.

3.3.5. Conclusions and Dissemination

The formulation of conclusions remains an essential part of the process, based on the results obtained from the evaluation of the proposed algorithms, as well as the analysis of the findings, highlighting the strengths and limitations of the approaches used. An additional contribution could include the identification of potential areas for improvement and recommendations for future research, as well as the dissemination of results and conclusions through more accessible means, such as infographics, blogs, or explanatory videos. This includes presenting results and conclusions at conferences and in specialized scientific journals to reach a broader audience and encourage the practical application of findings in the field of operations research and generalized vehicle routing. Additionally, feedback from the scientific community and peer review are of utmost importance.

4. Results and Discussion

The analysis of the results obtained from the heuristic algorithms and the proposed metaheuristic, the genetic algorithm, reveals important insights that contribute to enhancing their performance and achieving optimal solutions. The following discusses these results and highlights their impact on the optimization process.

4.1. Population Size Adjustment

During the executions, it was observed that increasing the population size was beneficial for improving the results. By increasing the number of individuals in the population, greater diversity of solutions and more comprehensive exploration of the search space were achieved. As a result, there was a significant reduction in the objective function value, indicating the identification of promising optimal solutions. This finding underscores the importance of considering a population large enough to obtain high-quality results.

4.2. Optimization of the Number of Generations

The number of generations is a key factor in the performance of the heuristic algorithms and the proposed genetic algorithm. As the number of iterations increases, a progressive refinement of the solutions is allowed. In the trials conducted, a clear trend was observed: as the number of generations increased, gradual improvements in the objective function value were achieved, approaching optimal solutions. This emphasizes the importance of considering a sufficient number of generations to permit convergence towards high-quality solutions.

4.3. Influence of Elite Count

The inclusion of elites in each generation proved to be an effective strategy for preserving high-quality solutions. By selecting the best individuals in each iteration, it ensures that their characteristics and advantages are transmitted to subsequent generations. During the executions, it was observed that increasing the number of elites led to an improvement in the results. This indicates that the best-found solutions were leveraged and refined, contributing to a faster convergence towards optimal solutions. The inclusion of elites guarantees that valuable characteristics are preserved and transmitted to following generations, thereby improving the quality of the obtained solutions.

4.4. Utilization of Advanced Operators

The incorporation of advanced crossover and mutation operators has significantly enhanced the performance of heuristic algorithms and the proposed genetic algorithm. These operators facilitate a more comprehensive exploration and diversification of potential solutions within the search space, leading to a considerable increase in result variability. This variability is crucial for identifying optimal solutions, reflecting a robust search strategy that effectively navigates complex landscapes. As a result, objective function values exhibit significant improvements, showcasing the effectiveness of these advanced methods in refining and enhancing solution quality. Moreover, these sophisticated operators promote a more strategic utilization of the search space, allowing for a deeper examination of solution combinations that traditional operators might overlook. This strategic exploration is essential for achieving higher quality and more efficient solutions in optimization. Ultimately, implementing advanced crossover and mutation techniques positions our algorithms to effectively address the complexities and challenges inherent in the Generalized Vehicle Routing Problem, resulting in more effective and sustainable routing solutions.

4.5. Improved Metaheuristic Algorithm

The introduction of the enhanced heuristic algorithm has yielded remarkable results in our experiments. This optimized methodology successfully integrates advanced search and optimization techniques, thereby demonstrating its effectiveness in achieving high-quality solutions. As shown in Table 1 and Figure 2, there is a notable reduction in the objective function value when employing this enhanced algorithm. This improvement significantly exceeds the results attained by the base model, underscoring the algorithm's superior performance and efficiency in optimization tasks.

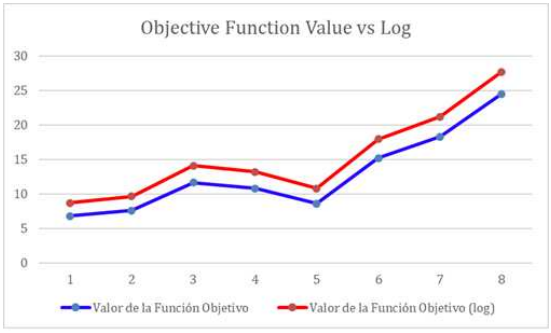
Table 1. *Results of Heuristic Algorithms for the Generalized Vehicle Routing Problem (GVRP)*

No. of Vehicles	Population Size	No. of Generations	Number of Elites	Best Solution Found	Objective Function Value	Objective Function Value (log)
2	50	10	10	[1 1 1 2 2]	6.82843	1.9224
2	100	15	20	[1 1 2 2 2]	7.65685	2.0327
3	200	20	30	[2 3 3 1 1]	11.6569	2.4611
3	300	25	40	[3 3 2 1 2]	10.8284	2.3833
3	50	10	10	[2 2 3 1 2]	8.65685	2.1597
4	100	15	20	[4 3 2 4 1]	15.2426	2.7241
5	200	20	30	[5 4 4 5 3]	18.3137	2.9034
6	300	25	40	[6 5 6 5 4]	24.5362	3.1984

Note: *This table provides performance metrics of heuristic algorithms applied to the Generalized Vehicle Routing Problem (GVRP).*

Table 1 illustrates a series of systematic improvements made during the execution trials, highlighting several key adjustments such as modifications to population size, the number of generations, and the retention of elite individuals, alongside the integration of advanced operators. The final row of the table presents the evaluation of the enhanced heuristic algorithm, showcasing the iterative refinement process that has been crucial to this research. Results from both the base model and the enhanced heuristic model emphasize the importance of strategic information selection and effective clustering techniques. These methodologies have significantly improved outcomes, as quantitatively demonstrated in Figure 3. This comparison not only emphasizes the superior performance of the enhanced algorithm but also validates the positive impact of these refinements in achieving higher-quality solutions throughout the optimization process.

Figure 3. *Comparison of objective function values and their logarithm for different configurations of the enhanced heuristic algorithm*



Note: *The figure compares the performance of various configurations of the Enhanced Heuristic Algorithm by displaying the objective function values and their logarithmic transformations, highlighting how different parameter settings impact solution quality.*

To further clarify the experimental methodologies, an additional column titled *configuration* was introduced, indicating the specific approach utilized in each experiment. Both the base configurations and proposed improvements relative to the base model were documented. Parameter adjustments were made to identify and select better configurations. The development of more advanced operators enabled the use of sophisticated selection, crossover, and mutation techniques, thereby enhancing the algorithm's performance. Additionally, an intensification of the number of elites was implemented. An alternative optimization heuristic, distinct from the original, was also improved, with the goal of exploring new approaches and discovering potentially superior solutions.

Table 2. *Experimental results of enhanced heuristic algorithm configurations*

Exp.	Configuration Type	No. of Vehicles	Population Size	No. of Generations	Quantity of Elites	Best Solution Found	Objective Function Value	Improvement over Base solution (%)
Base	Base Configuration	2	50	10	10	[1 1 1 2 2]	6.82843	-
Exp 1	Parameter setting	2	100	15	20	[1 1 2 2 2]	7.65685	12.14
Exp 2	Advanced operators	3	200	20	30	[2 3 3 1 2]	11.6569	70.88
Exp 3	More elites	3	300	25	40	[3 3 2 1 2]	10.8284	58.78
Exp 4	Improved heuristic algorithm	3	50	10	10	[2 2 3 1 2]	6.65685	26.83
Exp 5	Base Configuration	4	100	15	20	[4 3 2 4 1]	15.2426	122.23
Exp 6	Parameter setting	5	200	20	30	[5 4 4 5 3]	18.3137	168.16
Exp 7	Advanced operators	6	300	25	40	[6 5 6 5 4]	24.5362	259.96
Exp 8	More elites	4	150	25	10	[4 4 3 4 1]	13.9274	103.72
Exp 9	Improved heuristic algorithm	5	100	15	10	[5 5 5 5 2]	13.2426	94.14
Exp 10	Base Configuration	6	200	20	20	[6 6 6 6 3]	17.3137	153.47

Note: *This table summarizes the results of various experimental configurations applied to the Generalized Vehicle Routing Problem (GVRP) using a heuristic algorithm. The columns include the type of configuration, the number of vehicles used, the population size, the number of generations, and the number of elites selected during the optimization process.*

In Table 2, an additional column called *improvement over the base (%)* has also been added, indicating the percentage improvement in the objective function compared to the base configuration.

The choice of metric has allowed us to quantify the impact of each adjustment or approach on the improvement of results, as evidenced in Figure 4. The improved heuristics are capable of exploring the search space more efficiently and finding optimal solutions within a reasonable time. Additionally, we can note that the implementation of innovative strategies, particularly through adaptive heuristics, has led to a significant improvement in the objective function value when compared to the base model.

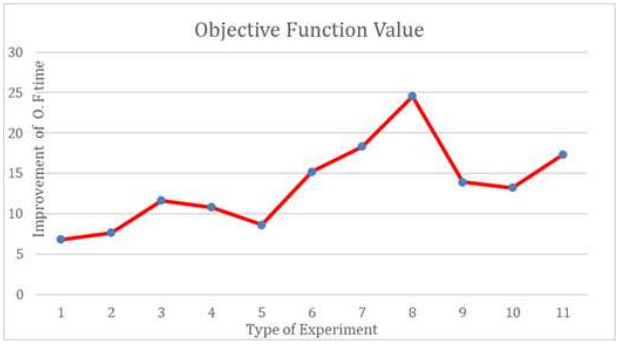


Figure 4. *Improvement over the base*

This figure 4 illustrates how different parameter settings and algorithm enhancements contribute to optimization. Percentage improvement in objective function values across various experimental configurations compared to the base model for the Generalized Vehicle Routing Problem (GVRP).

4.6. Advances in Computational Complexity and Optimization Techniques for the GVRP

The improved heuristic algorithm designed for the Generalized Vehicle Routing Problem (GVRP) exhibits computational complexity influenced by critical factors such as population size, the number of generations, and the computational load associated with the advanced operators employed in the selection, crossover, and mutation phases. This temporal complexity can be expressed using the notation

$$O(P \cdot G \cdot (C + O)),$$

where P represents the population size of the genetic algorithm, G the number of generations, C the complexity of the crossover operators, and O the complexity of the mutation operators. Such formalization is essential for a thorough analysis of each component's influence on the overall efficiency of the algorithm.

From the perspective of computational complexity theory, the GVRP is classified as an extension of the Vehicle Routing Problem (VRP), which is an NP-hard problem. This complexity arises from the additional constraints that must be considered, such as the assignment of customers to different subsets of vehicles, resulting in an exponential expansion of the search space. Exact formulations utilizing Mixed Integer Programming (MIP) face exponential complexities in the worst case, reaching $O(2^n)$ or worse, complicating their resolution in practical time frames for large instances.

In contrast, heuristic and meta-heuristic approaches, including Genetic Algorithms (GA), Ant Colony Optimization (ACO), and Particle Swarm Optimization (PSO), have proven capable of generating approximate solutions with significantly more manageable computational times. However, the effectiveness of these methods heavily depends on the balance achieved between exploration and exploitation. In this regard, our improved approach incorporates advanced crossover and mutation operators, as well as an intensification strategy that encompasses elite individual selection and dynamic parameter optimization, thereby achieving substantial improvements in algorithm efficiency without compromising solution quality.

Experimental results have indicated that these optimizations have led to notable reductions in objective function values, demonstrating improvements of up to 259.96% compared to baseline configurations, as detailed in Table 2. In light of comparisons with other meta-heuristic approaches, our method has shown greater stability in convergence towards high-quality solutions, thereby indicating a superior capacity for effectively exploring the solution space.

When contrasting these findings with alternative approaches, the Gomory Cut Plane Method is highlighted, which, although it guarantees optimal solutions, faces significant challenges due to its increasing exponential complexity, limiting its applicability to smaller instances. On the other hand, techniques such as Tabu Search (TS) and Simulated Annealing (SA) provide competitive solutions with lower computational requirements. However, these methods often struggle with the tendency to become trapped in local optima, especially in larger problems, thereby underscoring the need for flexible and robust approaches such as the one presented in this study.

5. Conclusions

This section outlines the key conclusions drawn from the progress of our research project on the Generalized Vehicle Routing Problem (GVRP), highlighting the advancements made and the significance of our findings.

5.1. Comprehensive Review and State of the Art

A thorough and unified review of all metrics utilized in the design of algorithms for the Traveling Salesman Problem (TSP), m-TSP, and the Heuristic Algorithms for Generalized Vehicle Routing Problem (AH-GVRP) was conducted. This analysis facilitated the identification of innovative methodologies and techniques that enhance the resolution of these complex vehicle routing problems. As a result, we achieved our objective of providing a comprehensive overview of the most effective strategies currently employed in this field.

5.2. Theoretical Foundations and Algorithm Development

The foundational theoretical results required for the design and implementation of heuristic and metaheuristic algorithms have been systematically structured. These approaches were pivotal in developing efficient algorithms for the AH-GVRP, effectively leveraging the heterogeneity of vehicle fleets while utilizing successive approximations and global search techniques. This enabled us to meet our goal of designing algorithms that significantly enhance goods distribution efficiency and minimize transportation costs.

5.3. Implementation and Evaluation of Algorithms

We developed and implemented functions and procedures for exact, heuristic, and metaheuristic algorithms using the C++ programming language, augmented by the commercial optimization libraries provided by IBM ILOG CPLEX Optimization Studio 12.6. This implementation facilitated a thorough assessment of the proposed Heuristic Algoritms for the Generalized Vehicle Routing Problem (AH-GVRP) algorithms compared to existing methodologies. Through extensive computational experiments and the application of pertinent performance metrics, we demonstrated the algorithms' ability to generate high-quality solutions and their effectiveness in practical applications.

5.4. Impact and Practical Applications

The major contribution of this research lies in the comparative analysis of our proposed models and the development of smaller-scale models for the GVRP. This evaluation confirmed the effectiveness of the AH-GVRP algorithms in real-world scenarios, as evidenced by their application to various case studies in route planning and vehicle allocation across different industries. The demonstrated positive impacts include notable reductions in transportation costs, optimization of resource utilization, and enhancements in service quality.

Acknowledgements

The authors thank the Department of Mathematics at Keiser University Latin American Campus and the Faculty of Science and Engineering at UNAN for their support. We also appreciate the valuable feedback from colleagues and the academic community. Special thanks to the funding organizations for enabling the computational experiments that validated our algorithms and advanced research on heuristic and metaheuristic approaches to the GVRP.

Conflicts of Interest

The authors declare no conflicts of interest. AI tools were used to improve the manuscript's language and clarity, with all content remaining the authors' sole responsibility.

References

[1] Ravindra A. et al. “Exact and heuristic algorithms for the weapon-target assignment problem”. In: *Operations research* 55.6 (2007), pp. 1136–1146. DOI: https://doi.org/10.1287/opre.1070.0440.

[2] John B. and Patrick M. “Ant colony optimization techniques for the vehicle routing problem”. In: *Advanced engineering informatics* 18.1 (2004), pp. 41–48. DOI: https://doi.org/10.1016/j.aei.2004.07.001.

[3] Roberto B., Enrico B., and Gilbert L. “Some applications of the generalized vehicle routing problem”. In: *Journal of the operational research society* 61.7 (2010), pp. 1072–1077.

[4] Tolga B., Güneş E., and Stefan R. “Formulations and branch-and-cut algorithms for the generalized vehicle routing problem”. In: *Transportation Science* 45.3 (2011), pp. 299–316. DOI: https://doi.org/10.1287/trsc.1100.0352.

[5] Thanh D. “Vehicle routing problems with additional constraints imposed on different groups of customers”. PhD thesis. Nantes, 2021.

[6] Vladimir D. and Zoran Š. “An efficient transformation of the generalized traveling salesman problem into the traveling salesman problem on digraphs”. In: *Information Sciences* 102.1-4 (1997), pp. 105–110. DOI: https://doi.org/10.1016/S0020-0255(96)00084-9.

[7] Asvin G. and Volker G. “A general vehicle routing problem”. In: *European Journal of Operational Research* 191.3 (2008), pp. 650–660.

[8] Bruce G. and Arjang A. “Vehicle routing: methods and studies”. In: *Studies in Management Science and Systems* 16 (1988).

[9] Gianpaolo G. and Gennaro I. “An efficient transformation of the generalized vehicle routing problem”. In: *European Journal of Operational Research* 122.1 (2000), pp. 11–17. DOI: https://doi.org/10.1016/S0377-2217(99)00073-9.

[10] Fernando H. and Wilmer L. “Algorithms, efficient implementations, and their utility in the industry: Generalized Vehicle Routing Problem (GVRP)”. In: *XXIV Simposio International de Métodos Matemáticos Aplicados a las Ciencias*. Universidad de Costa Rica. 2024, p. 65.

[11] Imdat K. and Tolga B. “Integer linear programming formulation of the generalized vehicle routing problem”. In: *Proc. of the 5-th EURO/INFORMS Joint International Meeting*. Citeseer Istanbul. 2003, July, pp. 06–10.

[12] Suresh K. and Ramasamy P. “A survey on the vehicle routing problem and its variants”. In: (2012).

[13] Ran L., Zhibin J., and Na G. “A hybrid genetic algorithm for the multi-depot open vehicle routing problem”. In: *OR spectrum* 36 (2014), pp. 401–421.

[14] Yiming L. et al. “Efficient feasibility checks and an adaptive large neighborhood search algorithm for the time-dependent green vehicle routing problem with time windows”. In: *European Journal of Operational Research* 310.1 (2023), pp. 133–155. DOI: https://doi.org/10.1016/j.ejor.2023.02.028.

[15] Silvia M. and Irene L. “An ant colony algorithm for the capacitated vehicle routing”. In: *Electronic Notes in Discrete Mathematics* 18 (2004), pp. 181–186. DOI: https://doi.org/10.1016/j.endm.2004.06.029.

[16] Petrică P., Oliviu M., and Corina S. “An improved hybrid algorithm for solving the generalized vehicle routing problem”. In: *Neurocomputing* 109 (2013), pp. 76–83. DOI: https://doi.org/10.1016/j.neucom.2012.03.032.

[17] Petrică P. et al. “Heuristic algorithms for solving the generalized vehicle routing problem”. In: *International Journal of Computers Communications & Control* 6.1 (2011), pp. 158–165.

[18] Puca P. et al. “A hybrid heuristic for a broad class of vehicle routing problems with heterogeneous fleet”. In: *Annals of Operations Research* 273 (2019), pp. 5–74. DOI: https://doi.org/10.1016/j.neucom.2012.03.032.

[19] Paolo T. and Daniele V. *The vehicle routing problem*. SIAM, 2002.

[20] Lan Z. and Dawei H. “Study on the vehicle routing problem considering congestion and emission factors”. In: *International Journal of Production Research* 57.19 (2019), pp. 6115–6129. DOI: https://doi.org/10.1080/00207543.2018.1533260.